

# CFD\_HW\_6

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课程：计算流体力学

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## 17. Lax–Wendroff Scheme

### (a) von Neumann Stability Analysis

The Lax–Wendroff scheme is:

$$u_j^{n+1} = u_j^n - \frac{a\Delta t}{2\Delta x}(u_{j+1}^n - u_{j-1}^n) + \frac{a^2\Delta t^2}{2\Delta x^2}(u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

Assume Fourier mode:

$$u_j^n = e^{\sigma n \Delta t} e^{ikj\Delta x}$$

Substitute into the scheme:

$$e^{\sigma \Delta t} = 1 - iC \sin \theta + C^2(\cos \theta - 1)$$

where:

$$C = \frac{a\Delta t}{\Delta x}, \quad \theta = k\Delta x$$

Magnitude:

$$|e^{\sigma \Delta t}|^2 = [1 + C^2(\cos \theta - 1)]^2 + (C \sin \theta)^2$$

Simplified result:

$$|e^{\sigma \Delta t}|^2 = 1 - C^2(1 - C^2) \sin^2 \theta \leq 1$$

Thus:

$$\boxed{|C| \leq 1}$$

## (b) Modified Equation

Time expansion:

$$u_j^{n+1} = u + \Delta t u_t + \frac{\Delta t^2}{2} u_{tt} + \frac{\Delta t^3}{6} u_{ttt} + \frac{\Delta t^4}{24} u_{tttt} + \dots$$

Space expansion:

$$u_{j\pm 1} = u \pm \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} \pm \frac{\Delta x^3}{6} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} \pm \dots$$

The modified equation:

$$u_t + a u_x = -\frac{\Delta t}{2} u_{tt} - \frac{\Delta t^2}{6} u_{ttt} - \frac{a \Delta x^2}{6} u_{xxx} + \frac{a^2 \Delta t}{2} u_{xx} + \frac{a^2 \Delta t \Delta x^2}{24} u_{xxxx} - \frac{\Delta t^3}{24} u_{tttt} + O(\Delta x^4, \Delta t^4)$$

Using:

$$u_{ttt} = -a^3 u_{xxx} \quad u_{tttt} = a^4 u_{xxxx}$$

Final modified equation:

$$u_t + a u_x = \frac{a(\Delta x)^2}{6} (C^2 - 1) u_{xxx} + \frac{a(\Delta x)^3}{8} C(1 - C^2) u_{xxxx} + \dots$$

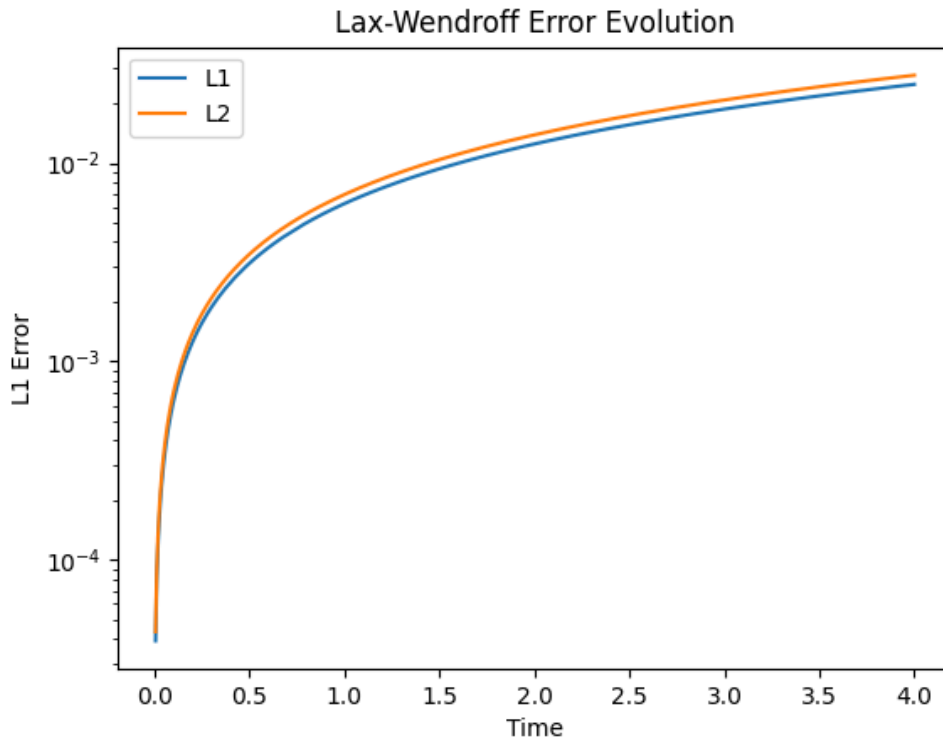
Coefficients:

$$C_1 = 0$$

$$C_2 = \frac{1}{6} (C^2 - 1)$$

$$C_3 = \frac{1}{8} C(1 - C^2)$$

## (c) Python Code for Error Evolution



## 18. Scheme B (Upwind Slope)

### (a) Stability Analysis

$$u_j^{n+1} = u_j^n - C(u_j^n - u_{j-1}^n) - \frac{C(1-C)}{2}(u_j^n - 2u_{j-1}^n + u_{j-2}^n)$$

Amplification factor:

$$e^{\sigma \Delta t} = 1 - C(1 - e^{-i\theta}) - \frac{C(1-C)}{2}(1 - e^{-i\theta})^2$$

Simplified result:

$$|e^{\sigma \Delta t}|^2 = 1 - C(1-C)^2(2-C)(1 - \cos \theta)^2 \leq 1$$

Stability condition:

$$|C| \leq 2$$

## (b) Modified Equation

Time expansion:

$$u_j^{n+1} = u + \Delta t u_t + \frac{\Delta t^2}{2} u_{tt} + \frac{\Delta t^3}{6} u_{ttt} + \frac{\Delta t^4}{24} u_{tttt} + \dots$$

Space expansion:

$$u_{j-1} = u - \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} - \frac{\Delta x^3}{6} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} - \dots$$

$$u_{j-2} = u - 2\Delta x u_x + \frac{(2\Delta x)^2}{2} u_{xx} - \frac{(2\Delta x)^3}{6} u_{xxx} + \frac{(2\Delta x)^4}{24} u_{xxxx} - \dots$$

The modified equation:

$$u_t + au_x = -\frac{\Delta t}{2} u_{tt} - \frac{\Delta t^2}{6} u_{ttt} - \frac{\Delta t^3}{24} u_{tttt} \\ + \frac{a\Delta x^2}{2} C u_{xx} + a\Delta x^2 \left( \frac{1}{3} - \frac{1}{2} C \right) u_{xxx} + a\Delta x^3 \left( -\frac{1}{4} - \frac{7}{24} C \right) u_{xxxx} + O(\Delta x^4, \Delta t^4)$$

Using:

$$u_{ttt} = -a^3 u_{xxx} \quad u_{tttt} = a^4 u_{xxxx}$$

Final modified equation

$$u_t + au_x = \frac{1}{6} (1 - C)(2 - C) u_{xxx} + \frac{1}{8} (1 - C)^2 (2 - C) u_{xxxx} + \dots$$

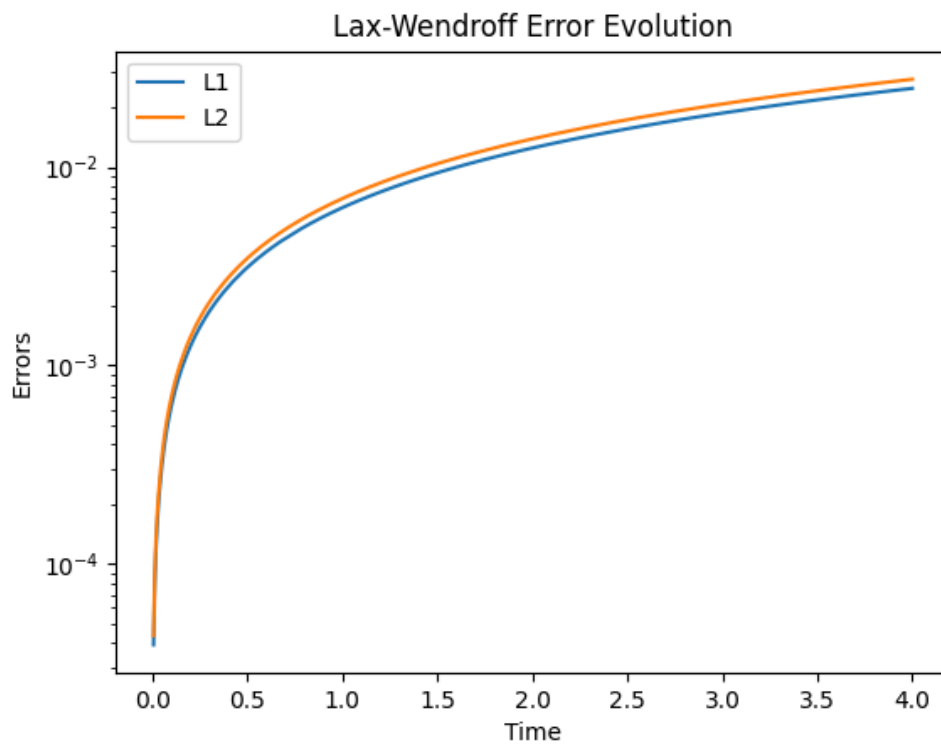
Coefficients:

$$C_1 = 0$$

$$C_2 = \frac{1}{6} (1 - C)(2 - C)$$

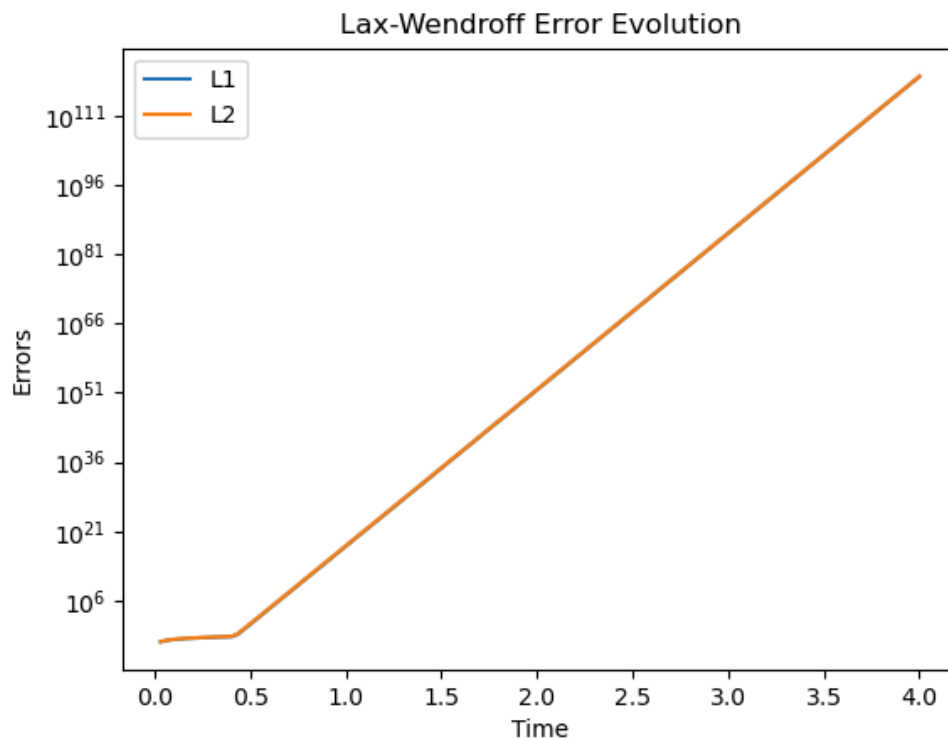
$$C_3 = \frac{1}{8} (1 - C)^2 (2 - C)$$

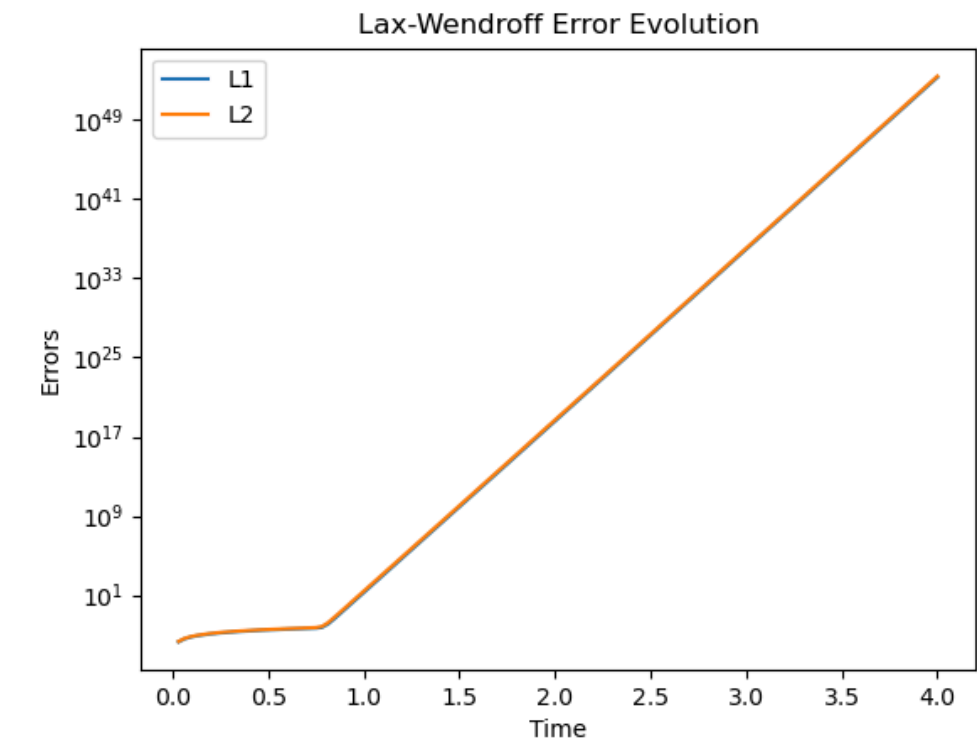
## (c) Python Code for Error Evolution



No clear different with 17.(C)

If we want the results like question 13, try change C to be 2.5:





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